

ISIC-2024 SLICE-3D — the honest frontier

Single-dataset · no external data · no synthetic positives

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The problem

- ISIC-2024 SLICE-3D: detect malignant skin lesions in 3D total-body-photography crops.
- 401,059 crops · 1,042 patients · 393 malignant / 400,666 benign → a 1,021:1 imbalance.
- Metric: **official pAUC above 80% TPR**, range [0, 0.20] (random $\approx$ 0.02, perfect = 0.20).
- The whole problem is the imbalance: 393 positives is *not much* for a deep net alone.

What we deliberately ban

- **No external data** — ISIC-2024 SLICE-3D only (no ISIC-2019/2020, no PAD-UFES, no external dermoscopy).
- **No generative / synthetic positives** — fabricating pathology breaks label validity.
- The winners' ~ 0.173 used **both**; matching it is **out of reach by construction**.

! Important

That impossibility *is the experiment*. So we change the question.

The question we actually answer

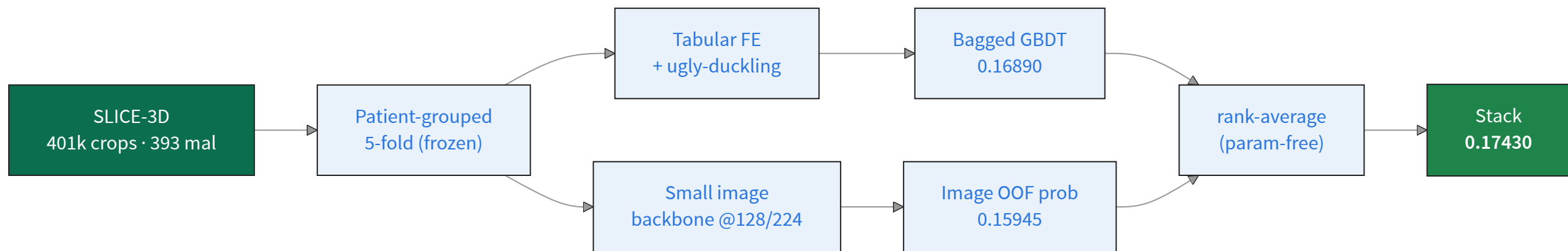
What is the best pAUC achievable per unit of inference cost — on SLICE-3D data alone?

- Not one private-LB number → a **quality-vs-cost Pareto frontier**.
- Axes: **pAUC** vs **params · FLOPs · CPU latency**.
- A model only counts if it **earns its cost**.

CV discipline — the whole game

- Leaky CV sent teams ~200 places down on the private split.
- Patient-grouped, target-stratified folds (5, SEED 42), frozen once into `data/folds.parquet`.
- No patient straddles folds. One metric, one implementation (`src/cv.py`), == official scorer.
- `cv-guardian` agent holds veto power over splits & metric.

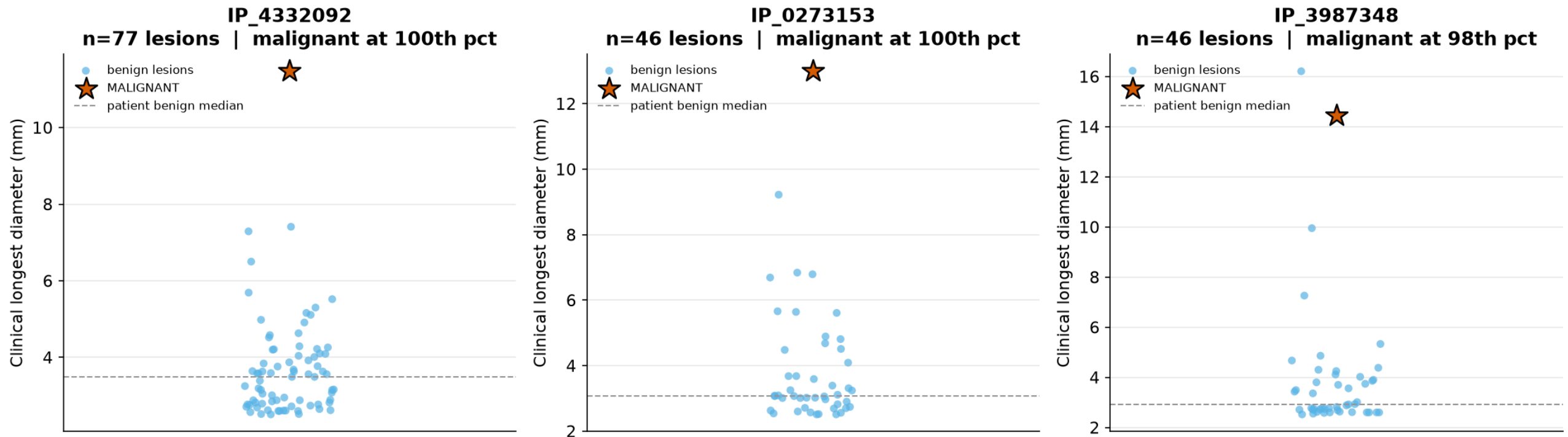
The pipeline



Where the signal lives

- The dominant signal is the **ugly-duckling sign**: how anomalous a lesion is vs the *same patient's* moles.
- Patient-relative features (`pdev_` / `prank_` / `pxc_`) = ~65% of GBDT gain.
- Top features: `tbp_lv_H` (hue, univariate AUC 0.81), `pdev_tbp_lv_H`, `pdev_clin_size`.

Ugly-duckling sign: within each patient the malignant lesion is a size outlier



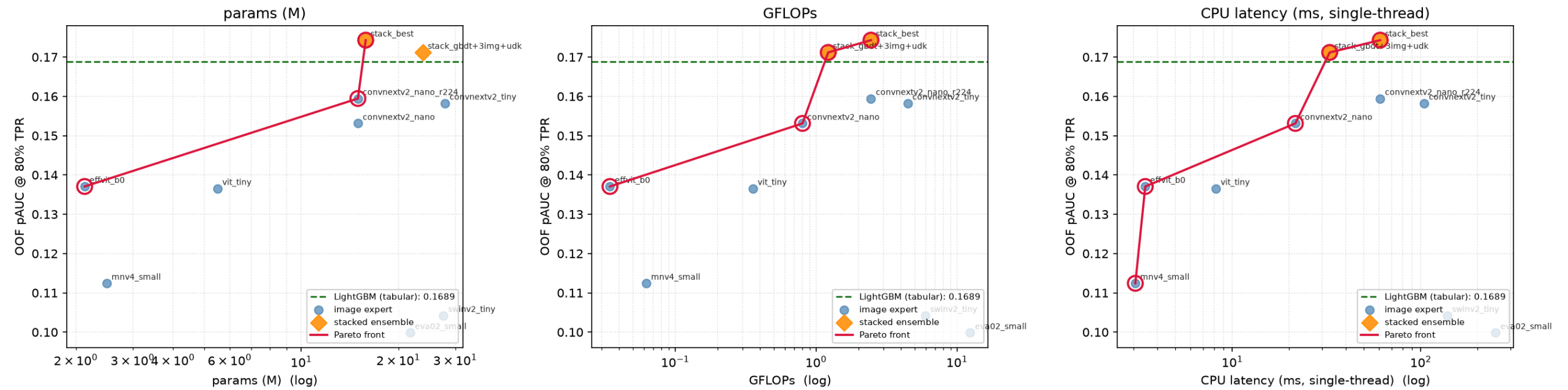
Headline results

Model	OOF pAUC@80%TPR	Params	GFLOPs	CPU ms
GBDT (tabular)	0.16890	0.86 M	~0	0.02
Best image (nano@224)	0.15945	14.98 M	2.46	60.9
Stack (rank-avg)	0.17430	15.84 M	2.46	60.9
Cheap stack	0.17117	23.84 M	1.22	32.9

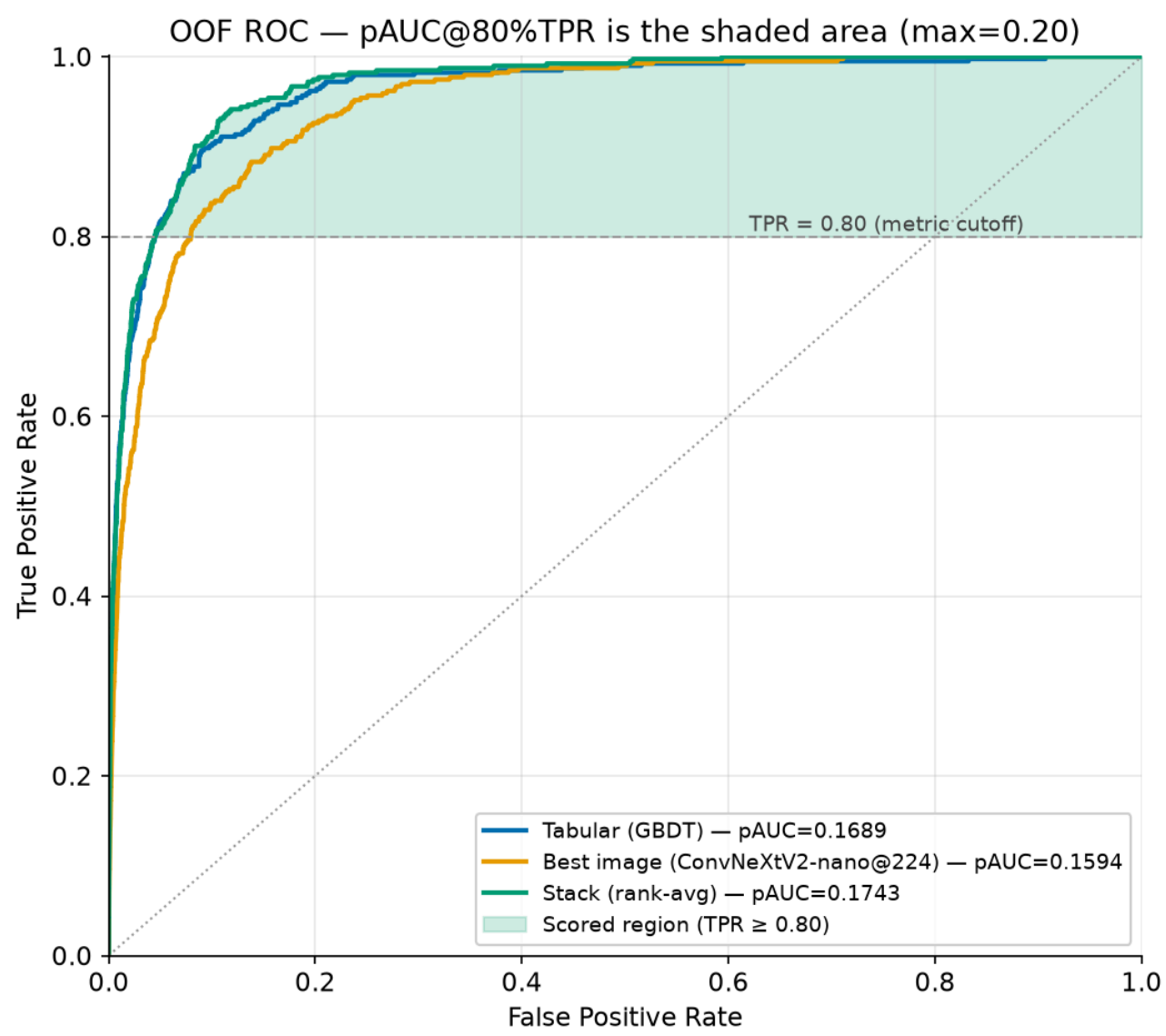
GBDT alone banks ~97% of the best score at ~free cost. The image expert earns **+0.0054** inside the stack.

The frontier

ISIC-2024 SLICE-3D quality-vs-cost frontier (authoritative cost; CPU latency single-thread; single-dataset, no external data)



The scored region (pAUC@80%TPR)



The stack sits above both experts **exactly where the metric integrates.**

Per-fold stability

	mean $\pm$ std
Tabular	0.1693 $\pm$ 0.0112
Best image	0.1597 $\pm$ 0.0085
Stack	0.1749 $\pm$ 0.0068

Fold variance (~0.025) > model gaps — yet the **stack wins/ties every fold** and has the lowest std.

Every complex idea lost

Negative results — complexity did not pay off (all OOF, leak-verified)

Idea tried	Its pAUC	Beaten by	Honest finding
Learned per-lesion gate (MoE)	0.15007	0.16890 (tabular)	LOSES to tabular alone
Meta-LGBM stacker	0.17108	0.17430 (rank-avg)	loses to trivial rank-avg
+ PCA image embeddings into GBDT	< 0.1743	0.17430	embeddings HURT
Heavy backbones (swinv2 / eva02)	0.104 / 0.100	0.15945 (nano@224)	dominated — overfit @393 pos
EMA 0.999 (over-smoothing)	~0.146	0.15945 (EMA0.995)	stronger EMA hurts

- Learned MoE gate **0.150** — *below tabular alone*.
- Meta-LGBM stacker 0.171 < trivial 0.174. Image embeddings: hurt. Heavy transformers: collapse.

Why simpler won

- Every learned add-on must fit parameters from the **same 393 positives** the experts already used.
- A rank-average has **zero parameters** → cannot overfit the folds.
- At extreme imbalance: **bias toward zero-parameter fusion + small backbones** — *also* the cheapest.

Credit where due

- **1st** — Ilya Novoselskiy (EVA-02 + EdgeNeXt + GBDT, private 0.17264; external + ~30k synthetic — both banned here; synthetic added only **+0.0007** in their own ablation).
- **2nd** — uchiyama33, **3rd** — kyohei-123.
- Public lineage: greysky (tabular), snnclsr, motono0223, **andreasbis** (CNN-preds-as-feature), richolson.

Conclusion

- Defensible claim: **SOTA among single-dataset, no-external-data, no-synthetic solutions, on a quality–efficiency frontier.**
- Best constrained point: **OOF pAUC 0.17430 @ 15.8 M / 2.46 GFLOPs / 61 ms CPU.**
- Validation discipline + an explicit cost axis give the result teeth.
- Reproducible: [SEED=42](#), config-driven, [make folds](#) → [gbdt](#) → [vision](#) → [frontier](#) → [site](#).

Thank you — Pakistan.AI

- Raja, Muhammad Junaid Ali Asif — M11217073
- Sultan, Adil — M11217078
- Hassan, Shahzaib Ahmed — M11217081
- Neural Networks · FIC Lab / CSIE, NYUST
- Full report, frontier, and reproduction recipe on the companion site.